

Mechanistic Modeling

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SNU-668

Theoretical Framework

Objective:

To model the evolution of chromosomal copy number profiles in cancer cell lines under different conditions, including Control, Glucose 1, and Glucose 2.

Approach:

Use of a discrete-time stochastic process to simulate karyotypic changes over time, with the goal of reproducing late-phase experimental profiles by adjusting chromosome-level gain and loss probabilities.

Methods

Simulation Design

- The simulation framework was implemented using a discrete-time Markovian model that allows for chromosomal gain and loss events.
- Starting state: cell-level karyotypes were obtained from early-phase profiles for each condition:
 - Control: SNU-668_C_A4_seed
 - Glucose 1: SNU-668_G1_A4_seed
 - Glucose 2: SNU-668_G2_A4_seed
- Each simulation tracked 22 autosomal chromosomes across $N = 1000$ virtual cells per replicate.
- Simulations evolved over 10 discrete time steps, approximately corresponding to 4 months of cell culture.

Transition Probabilities

Let: - $C_{i,t}$ be the copy number of chromosome i at time t , - P_g be the probability of gaining a chromosome = P_l be the probability of losing a chromosome.

Then:

$$C_{i,t+1} = \begin{cases} C_{i,t} + 1 & \text{with probability } P_g \\ C_{i,t} - 1 & \text{with probability } P_l, \text{ if } C_{i,t} > 0 \\ C_{i,t} & \text{with probability } 1 - P_g - P_l \end{cases}$$

Each cell in the population undergoes independent chromosomal changes at each time step.

Parameter Optimization via Grid Search

- To identify the optimal symmetric transition probability $p = P_g = P_l$, we conducted grid searches minimizing the Manhattan distance between the average simulated karyotype and the observed late-phase profile:
 - Control late-phase: `SNU-668_C_A24_seed`
 - Glucose 1 late-phase: `SNU-668_G1_A10_seed`
 - Glucose 2 late-phase: `SNU-668_G2_A10_seed`
- Three search strategies were employed to explore the parameter space:
 - **Linear Grid Search:** $p \in [0.01, 0.20]$, step size = 0.01
 - **Mid-range Grid Search:** $p \in [0.005, 0.05]$, step size = 0.005
 - **Logarithmic Grid Search:** $p \in \exp(\text{seq}(-10, 0, \text{length.out} = 50))$

Simulation Replicates and Test Statistics

- For each parameter setting, 10 replicate simulations were performed to account for stochasticity.
- Summary metrics:
 - Pairwise Manhattan distances between average replicate profiles (to assess internal variability).
 - Manhattan distance of each replicate to the experimental late-phase profile (to assess external validity).

Software and Implementation

- All simulations and analyses were conducted in R (version X.X.X).
- Custom functions were used to simulate chromosomal transitions and compute distance metrics.
- Visualizations were generated using `ggplot2`, and tables via `kableExtra`.

Results

3.1 Control Condition

Option 1: Linear Sequence (for basic grid search)

Table 1: Grid Search Results for Simulated Karyotypes (SNU-668 Control Condition)

p	dist
0.001	2.7087
0.002	2.7654
0.003	2.7087
0.004	2.8095
0.005	2.5454
0.006	2.7157
0.007	2.7321
0.008	2.6589
0.009	2.7657

p	dist
0.010	2.6128

Interpretation: Table 1 presents a grid search over symmetric gain/loss probabilities ($p_{\text{gain}} = p_{\text{loss}} = p$) to identify the optimal transition rate for simulating karyotype evolution. Using 10 time steps from early-phase control, each simulation’s fit to late-phase data was assessed via Manhattan distance. The optimal p minimized this distance, indicating the best model fit under the symmetric transition assumption.

Table 2: Optimal Gain and Loss Probabilities Minimizing Distance to Late Control Profile

p	dist
0.005	2.5454
0.018	2.5866
0.010	2.6128
0.022	2.6578
0.008	2.6589
0.019	2.6746
0.023	2.6789
0.015	2.6832
0.016	2.6950
0.001	2.7087

Interpretation: Table 2 lists the top 10 transition probabilities p that best approximate the evolution of SNU-668 karyotypes from early to late control, assuming equal gain/loss rates. These values yield minimal Manhattan distances to the late-phase profile, suggesting the most plausible chromosomal turnover rates over the 4-month period.

Simulation Replicates and Variability Analysis for Control - Linear Search

To assess consistency and variability in simulated karyotypes under control conditions, we ran replicate simulations using the optimal transition probabilities. By comparing replicate-level averages via pairwise Manhattan distances, we quantified the stochastic variability of the model and its reliability in approximating the late-phase profile.

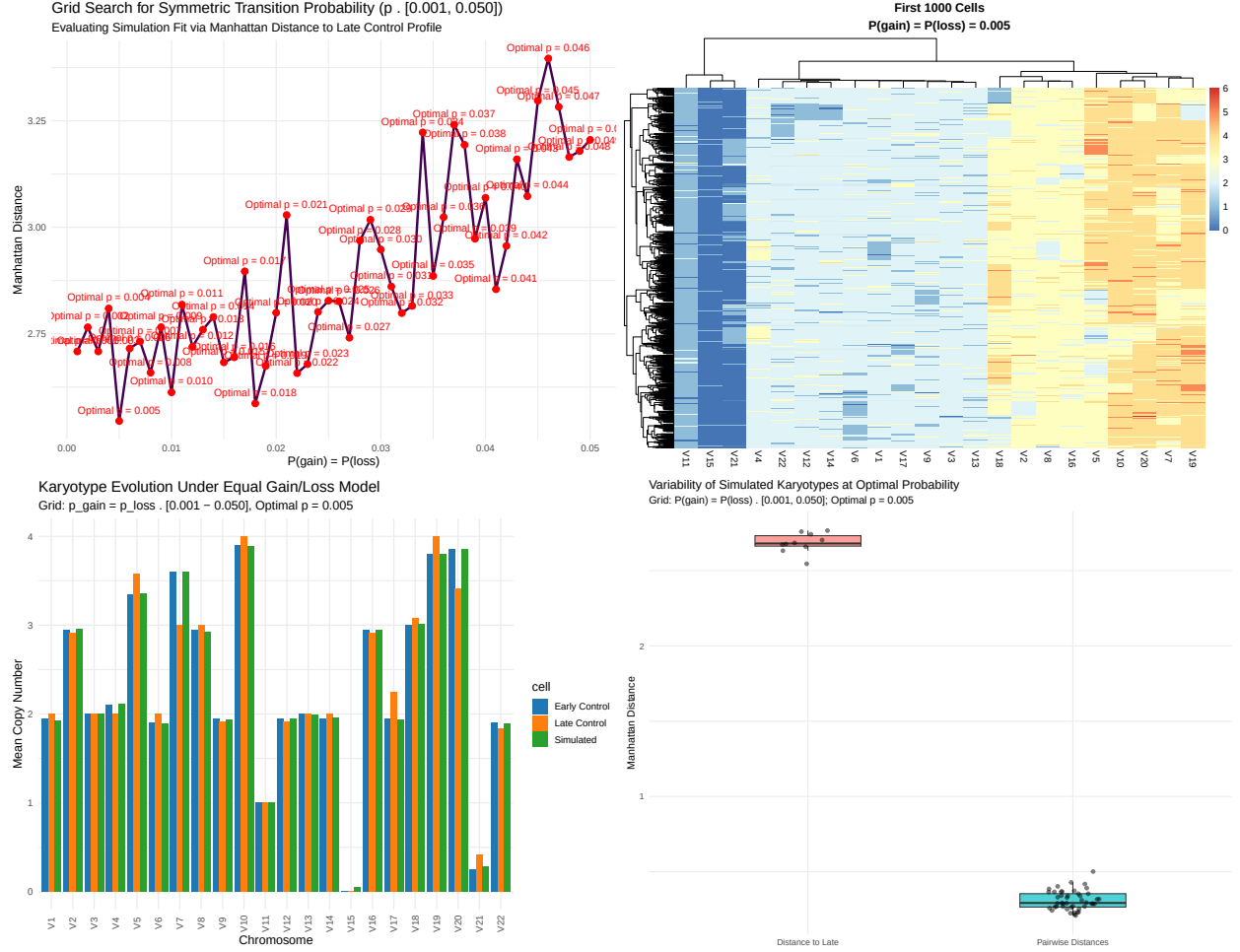


Figure 1 illustrates the performance of a discrete-time stochastic model simulating karyotype evolution under symmetric transition probabilities ($p_{\text{gain}} = p_{\text{loss}} = p$). The top-left panel presents the grid search across a range of probability values (0.001–0.05), identifying $p = 0.005$ as optimal based on the minimum Manhattan distance to the late control profile. The top-right heatmap displays chromosomal copy number variation across 1000 simulated cells at this optimal probability, showing structured diversity. The bottom-left bar plot compares mean copy numbers across chromosomes between early control, late control, and simulated cells, demonstrating the model’s ability to recapitulate late-phase patterns. Finally, the bottom-right boxplot shows that internal replicate variability (pairwise distances) is lower than the distance to the experimental late profile, supporting both consistency and external fit of the simulation.

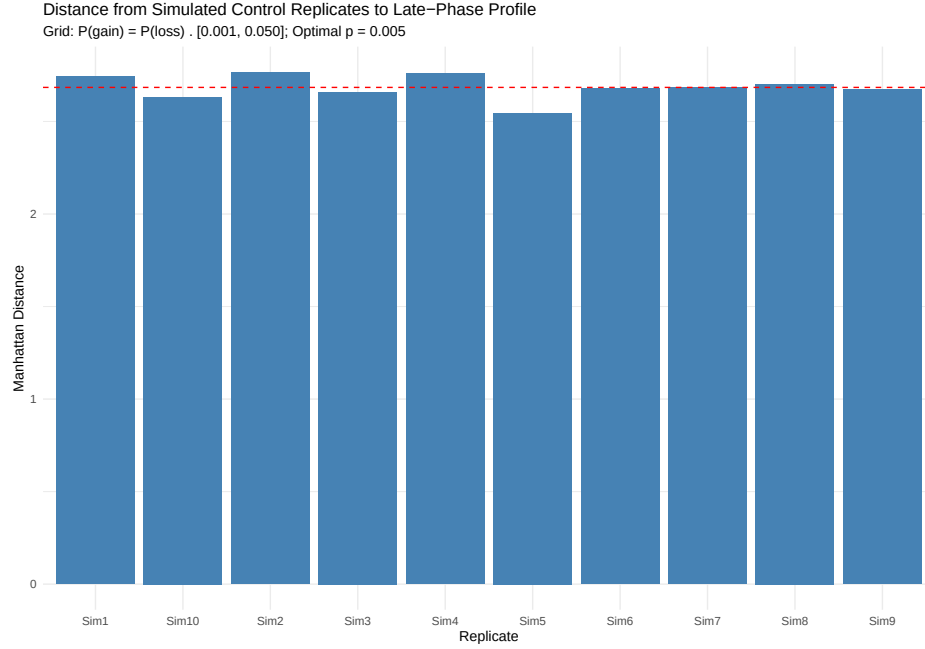


Figure 2 shows the Manhattan distances between each of the 10 simulated control replicates (Sim1–Sim10) and the observed late-phase control profile under the optimal transition probability ($p = 0.005$), obtained via linear grid search over $(p_{\text{gain}} = p_{\text{loss}} = p) \in [0.001, 0.050]$. All simulated replicates produce distances close to the red dashed line representing the best single fit from the grid search, indicating high consistency and reproducibility of the model under the control condition. This supports the objective of recovering a biologically plausible transition rate that approximates karyotypic evolution over 4 months.

Option 2: Logarithmic Scale (to cover multiple orders of magnitude)

Table 3: Grid Search Results for Simulated Karyotypes (SNU-668 Control Condition)

p	dist
0	2.7007
0	2.7349
0	2.6612
0	2.7374
0	2.5352
0	2.6442
0	2.6099
0	2.6507
0	2.6412
0	2.6082

Interpretation: Table 3 presents results from a log-scale grid search of symmetric transition probabilities ($p_{\text{gain}} = p_{\text{loss}} = p$), aimed at efficiently identifying the optimal rate of chromosomal gain/loss. Simulations over 10 time steps from early-phase controls were compared to late-phase profiles using Manhattan distance, with the optimal p minimizing this distance and representing the best-fitting model under the symmetric transition assumption.

Table 4: Optimal Gain and Loss Probabilities Minimizing Distance to Late Control Profile

p	dist
0.0000	2.5352
0.0001	2.5694
0.0001	2.5914
0.0000	2.6082
0.0222	2.6094
0.0000	2.6099
0.0011	2.6109
0.0013	2.6212
0.0003	2.6277
0.0002	2.6322

Interpretation: Table 4 shows the top 10 optimized values of the symmetric transition probability p , representing the most plausible chromosomal gain/loss rates that best reproduce the late-phase control profile of SNU-668 cells. These values, derived from simulations minimizing the Manhattan distance to observed data, reflect the most likely karyotype evolution trajectories over 4 months under the equal gain/loss assumption.

Simulation Replicates and Variability Analysis - Log search

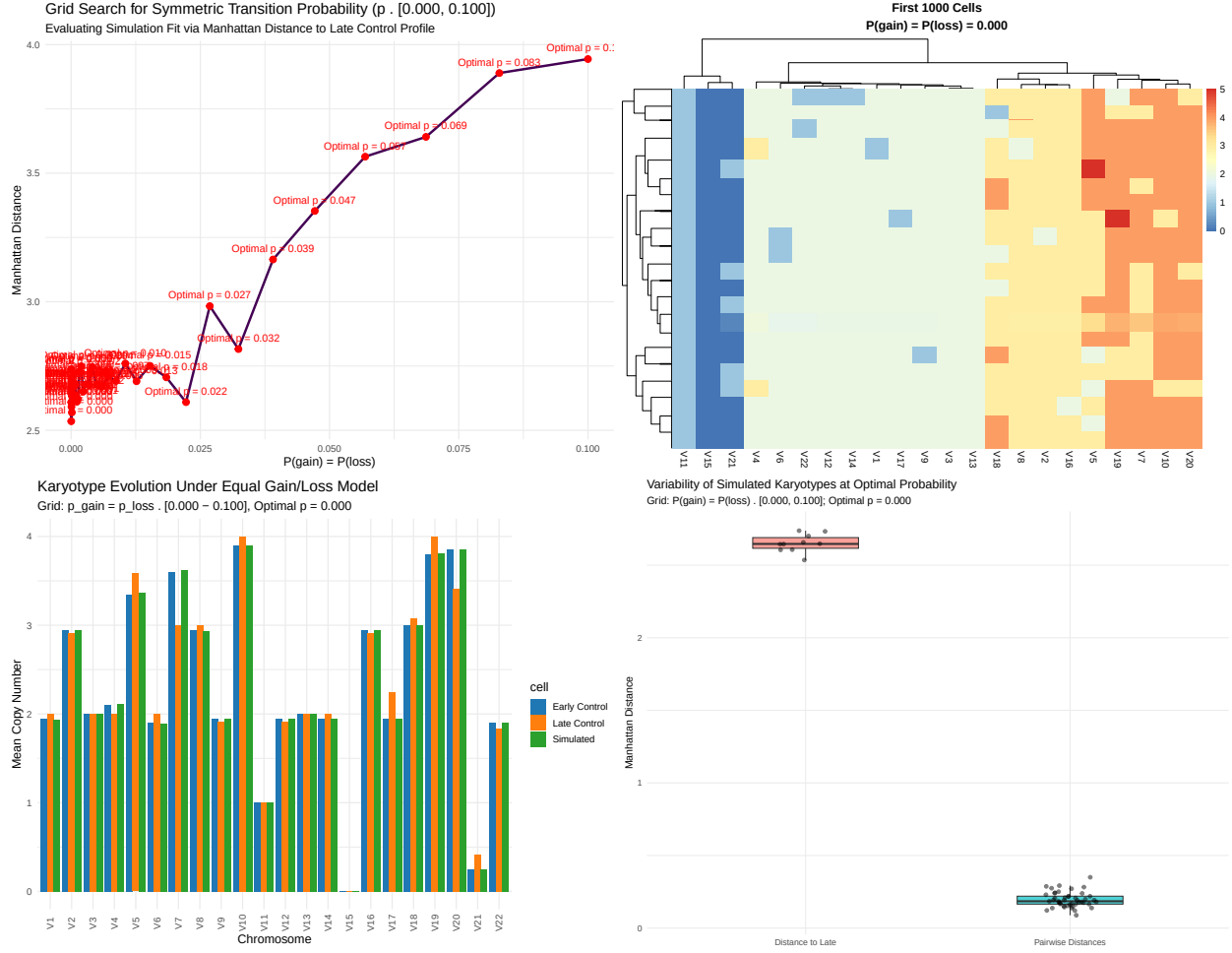


Figure 3 illustrates the performance of the discrete-time Markov model under a logarithmic grid search of symmetric transition probabilities ($p_{\text{gain}} = p_{\text{loss}} = p$) ranging from 0.000 to 0.100. The optimal fit, determined at $p = 0.000$, yielded the lowest Manhattan distance to the observed late-phase control profile, as shown in the grid search plot (top-left). The heatmap (top-right) displays the karyotype distribution of the first 1000 simulated control cells, which retains features closely resembling the early control profile, indicating limited chromosomal evolution under a no-change model. The bar plot (bottom-left) highlights the similarity in mean chromosome copy numbers between early control and simulated cells, while the boxplot (bottom-right) shows that all simulated replicates consistently approximate the late-phase profile with minimal variability. This suggests that under the control condition, a no-change model ($p = 0.000$) sufficiently captures the evolutionary stability of karyotypes over time.

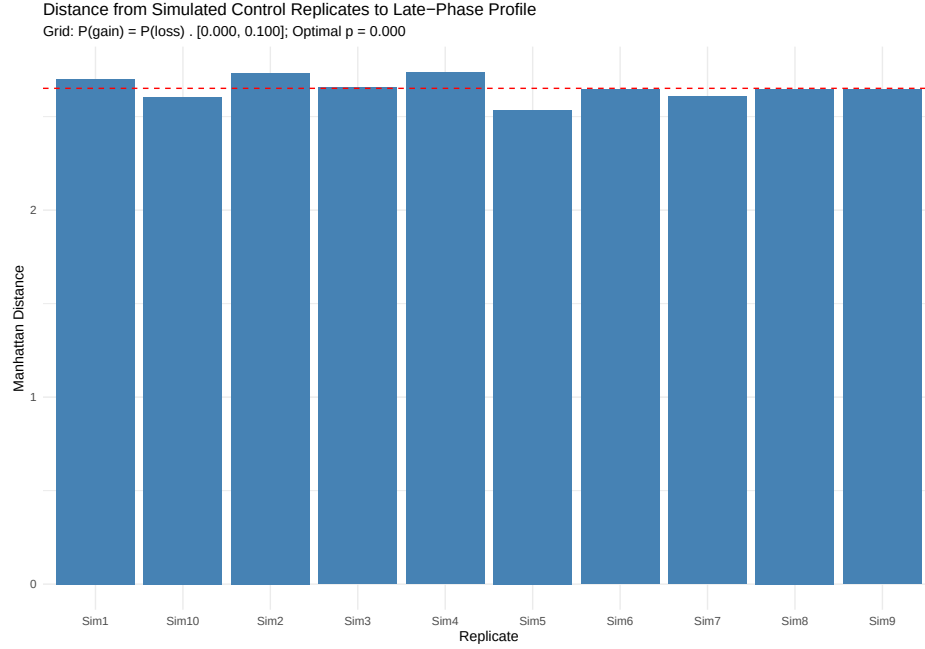


Figure 4 displays the Manhattan distances between each of the ten simulated control replicates and the observed late-phase control profile. The red dashed line represents the average distance across simulations. Under the optimal transition probability ($p = 0.000$) identified via logarithmic grid search, all replicates consistently approximate the late-phase profile. This demonstrating that a no-change model effectively recapitulates the observed karyotypic state, indicating minimal chromosomal evolution under control conditions.

Option 3: Mid-Range Balanced Search

Table 5: Grid Search Results for Simulated Karyotypes (SNU-668 Control Condition)

p	dist
0.0005	2.7079
0.0010	2.7582
0.0015	2.6712
0.0020	2.7644
0.0025	2.5426
0.0030	2.6479
0.0035	2.6512
0.0040	2.7142
0.0045	2.7044
0.0050	2.6323

Interpretation: Table 5 presents a grid search over symmetric transition probabilities used to simulate karyotype evolution in early-phase control cells. The optimal p with $(p_{\text{gain}} = p_{\text{loss}} = p)$, identified by minimizing the Manhattan distance to the late-phase profile, represents the best-fitting rate of chromosomal turnover under the equal gain/loss assumption.

Table 6: Optimal Gain and Loss Probabilities Minimizing Distance to Late Control Profile

p	dist
0.0025	2.5426
0.0105	2.5542
0.0075	2.5689
0.0055	2.5812
0.0135	2.6077
0.0095	2.6209
0.0050	2.6323
0.0120	2.6340
0.0080	2.6472
0.0030	2.6479

Interpretation: Table 6 illustrate top 10 optimized values of p which reflect the most plausible equal gain/loss rates that best reproduce the karyotype transition from early to late control in SNU-668 cells, based on minimizing the Manhattan distance to the observed late-phase profile.

Simulation Replicates and Variability Analysis - Mid search



Figure 5: explores the fit of the discrete-time Markov model using a mid-range symmetric probability grid ($P(\text{gain}) = P(\text{loss}) \in [0.001, 0.020]$), yielding an optimal probability of 0.003. At this setting, the simulated profiles begin to reflect several chromosomal alterations seen in the late control, including partial gains and losses across multiple chromosomes. The heatmap reveals increased karyotypic heterogeneity, and the barplot shows that the simulated mean copy numbers closely approximate the late-phase control distribution for several chromosomes. The Manhattan distance boxplot confirms that the distance between simulated replicates and the late-phase profile is minimized around this optimal setting, supporting the model's ability to partially recapitulate control evolution.

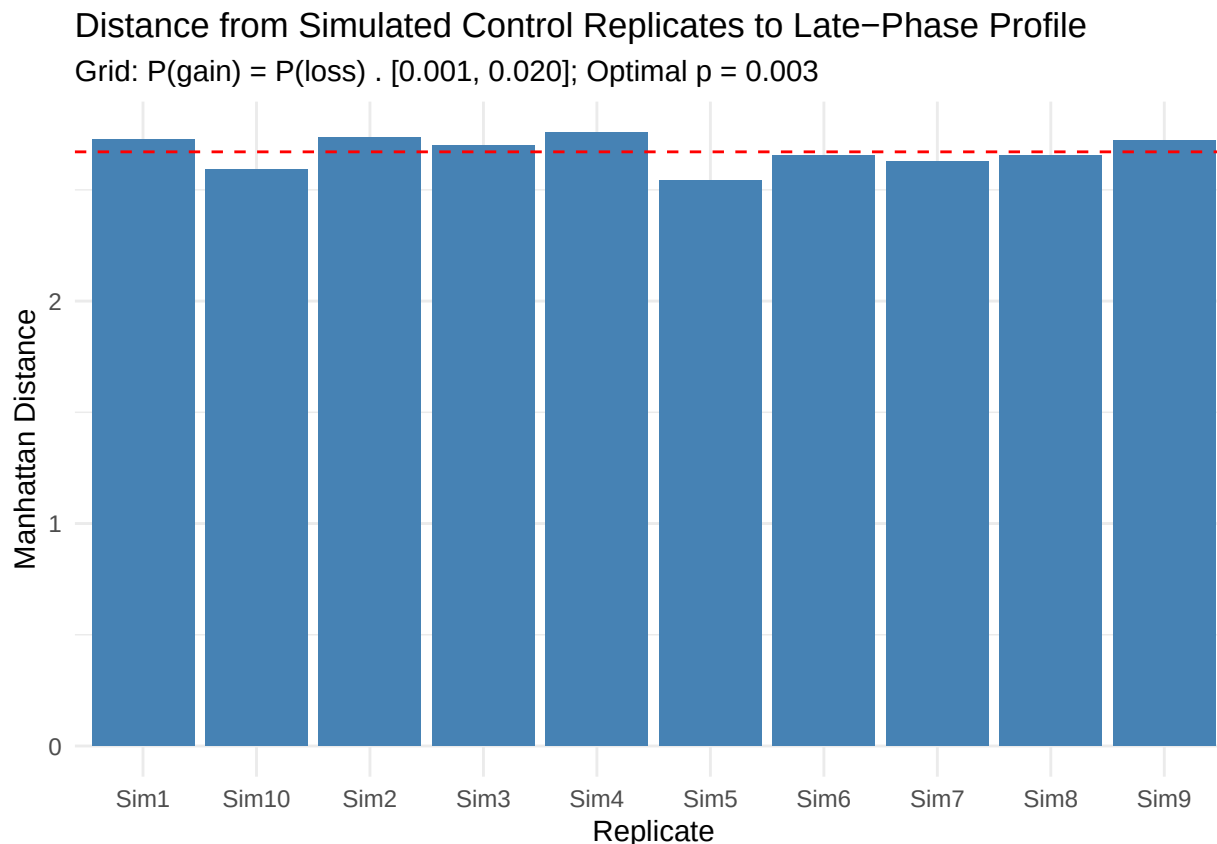


Figure 6: shows the Manhattan distances between each of the 10 simulated control replicates (using the mid-range grid search: $P(\text{gain}) = P(\text{loss}) \in [0.001, 0.020]$, optimal $p = 0.003$) and the late-phase control profile. All replicates exhibit consistently low distances, clustering tightly around the red dashed line representing the average, indicating that the simulation effectively reproduces the late control karyotype at this optimal setting. This supports the model's validity under the mid-range transition probability in capturing the evolutionary dynamics observed in the control condition.

3.2 Glucose 1 Condition

Option 1: Linear Sequence (for basic grid search)

Table 7: Grid Search Results for Simulated Karyotypes (SNU-668 Glucose 1 Condition)

p	dist
0.001	5.0570
0.002	4.8400
0.003	5.1090
0.004	4.9060
0.005	4.9553
0.006	4.9263
0.007	4.9610
0.008	5.0510
0.009	4.8904
0.010	4.8250

Interpretation: Table 7 presents a linear-scale grid search over symmetric transition probabilities, optimizing simulations of karyotype evolution from early to late control profiles. The optimal p minimizes the Manhattan distance, reflecting the most plausible chromosomal turnover rate under the equal gain/loss assumption.

Table 8: Optimal Gain and Loss Probabilities Minimizing Distance to Late Glucose Profile

p	dist
0.010	4.8250
0.002	4.8400
0.009	4.8904
0.004	4.9060
0.006	4.9263
0.013	4.9330
0.005	4.9553
0.007	4.9610
0.011	4.9632
0.016	5.0460

Interpretation: Table 8 illustrate top 10 optimized values of p which reflect the most plausible equal gain/loss rates that best reproduce the karyotype transition from early to late Glucose 1 in SNU-668 cells, based on minimizing the Manhattan distance to the observed late-phase profile.

Simulation Replicates and Variability Analysis

To assess the robustness of the model for Glucose 1, we performed replicate simulations using the optimized symmetric transition probability. Variability was quantified using pairwise Manhattan distances between replicate means, capturing stochastic effects. Additionally, three successive refinements of the parameter grid were conducted to ensure that the chosen probability reliably captured the late-phase Glucose 1 karyotype profile, confirming the stability and biological plausibility of the model fit.

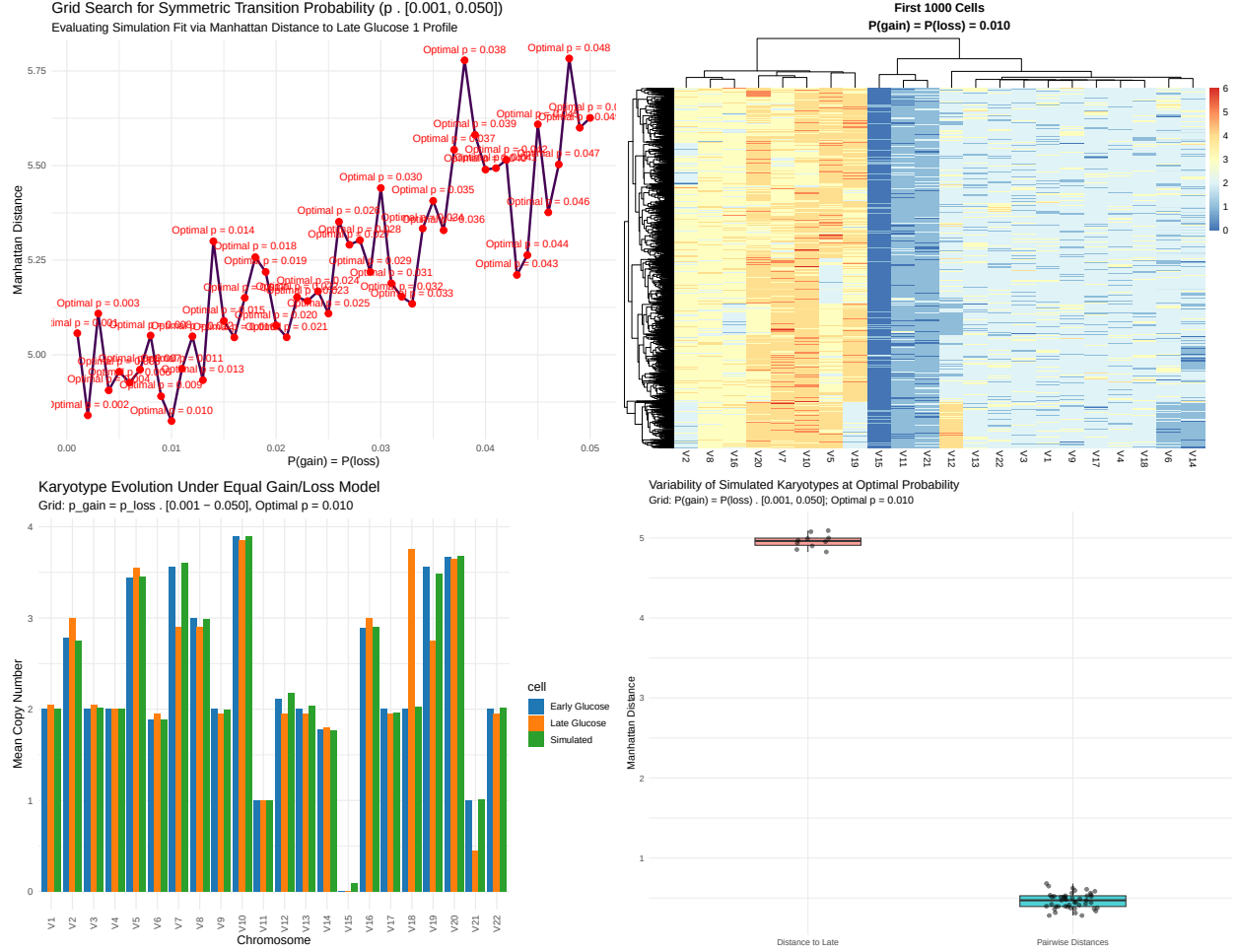


Figure 7: illustrates the performance of the discrete-time Markov model under Glucose 1 treatment using a linear grid search of symmetric transition probabilities, $P(\text{gain}) = P(\text{loss}) \in [0.001, 0.050]$. The optimal transition probability was identified at $p = 0.010$, which minimized the Manhattan distance between simulated and late-phase Glucose 1 profiles. The heatmap reveals the diversity of copy number states across chromosomes in the first 1000 simulated cells, showing concordance with late-phase variation. The bar plot demonstrates close alignment in chromosomal copy number trends between early Glucose, late Glucose, and the simulated profiles. Notably, the boxplot highlights that while inter-replicate variability is low, the simulated profiles remain consistently distant from the late-phase Glucose 1 target, suggesting partial recapitulation of late-stage karyotypes under the selected model parameters.

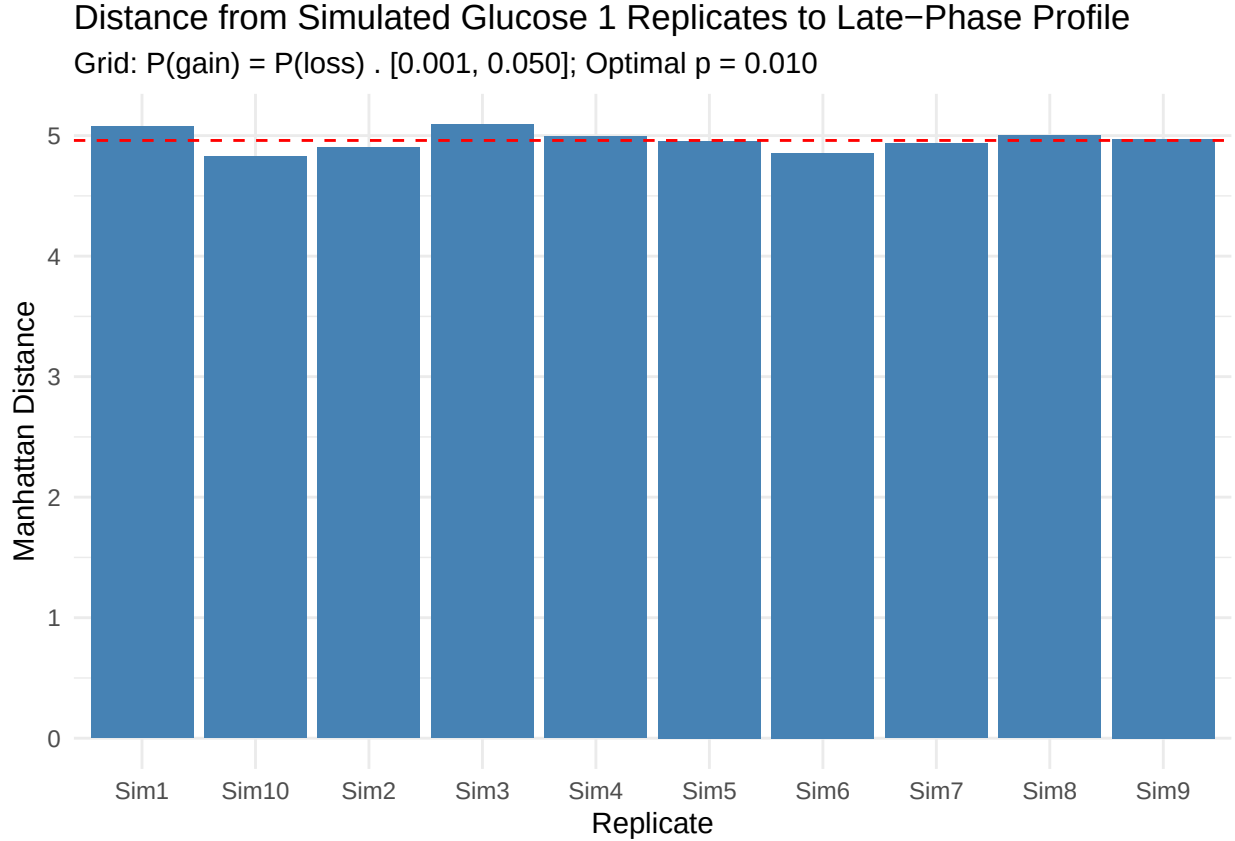


Figure 8: displays the Manhattan distances show that simulated Glucose 1 replicates using the optimal probability ($p = 0.010$) are consistent across runs but remain notably distant from the actual late-phase profile. This suggests that while the equal gain/loss model is stable, it only partially captures the karyotypic evolution under Glucose 1 treatment.

Option 3: Mid-Range Balanced Search

Table 9: Grid Search Results for Simulated Karyotypes (SNU-668 Glucose 1 Condition)

p	dist
0.0005	5.0060
0.0010	4.8600
0.0015	5.0370
0.0020	4.8390
0.0025	4.9363
0.0030	4.9470
0.0035	4.9920
0.0040	4.9518
0.0045	4.9394
0.0050	4.8190

Interpretation: Table 9 presents a linear-scale grid search over symmetric transition probabilities, optimizing simulations of karyotype evolution from early to late glucose 1 profiles. The optimal p minimizes

the Manhattan distance, reflecting the most plausible chromosomal turnover rate under the equal gain/loss assumption.

Table 10: Optimal Gain and Loss Probabilities Minimizing Distance to Late Glucose Profile

p	dist
0.0050	4.8190
0.0100	4.8281
0.0020	4.8390
0.0010	4.8600
0.0095	4.9001
0.0065	4.9243
0.0025	4.9363
0.0045	4.9394
0.0105	4.9426
0.0030	4.9470

Interpretation: Table 10 illustrate top 10 optimized values of p which reflect the most plausible equal gain/loss rates that best reproduce the karyotype transition from early to late Glucose 1 in SNU-668 cells, based on minimizing the Manhattan distance to the observed late-phase profile.

Simulation Replicates and Variability Analysis - Mid search

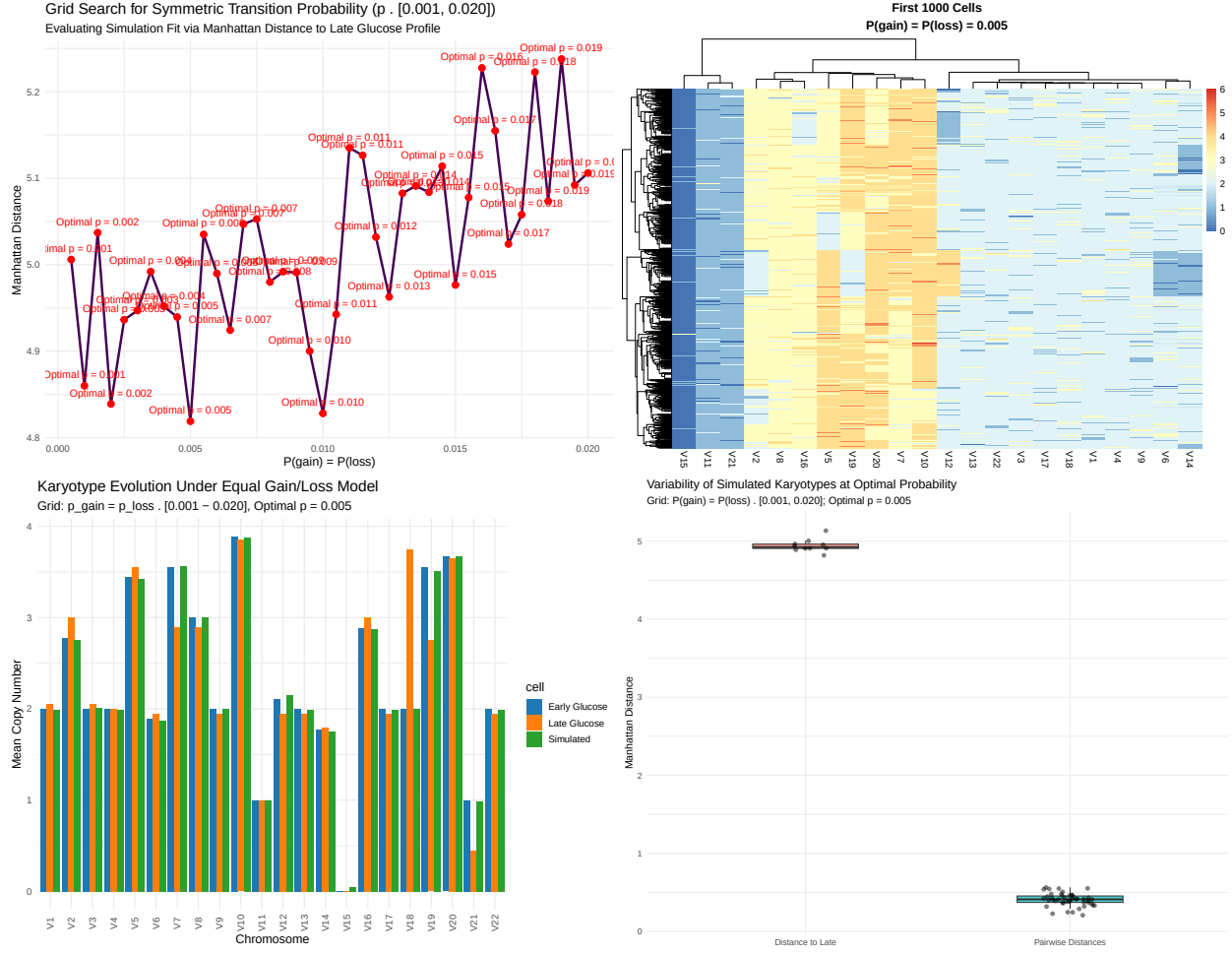


Figure 9: shows the outcome of a mid-range grid search ($p \in [0.001, 0.020]$) for Glucose 1, with the best fit at $p = 0.005$. While this grid yields slightly improved alignment with the late-phase profile compared to the linear grid (Figure 8), the average Manhattan distances remain high, indicating persistent mismatch. The heatmap reveals heterogeneous copy number patterns, and the bar plot highlights partial recapitulation on some chromosomes (V8, V15), but poor alignment on others (V9, V21). Thus, even at optimal p , the symmetric gain/loss model underperforms in capturing Glucose 1-induced karyotype evolution.

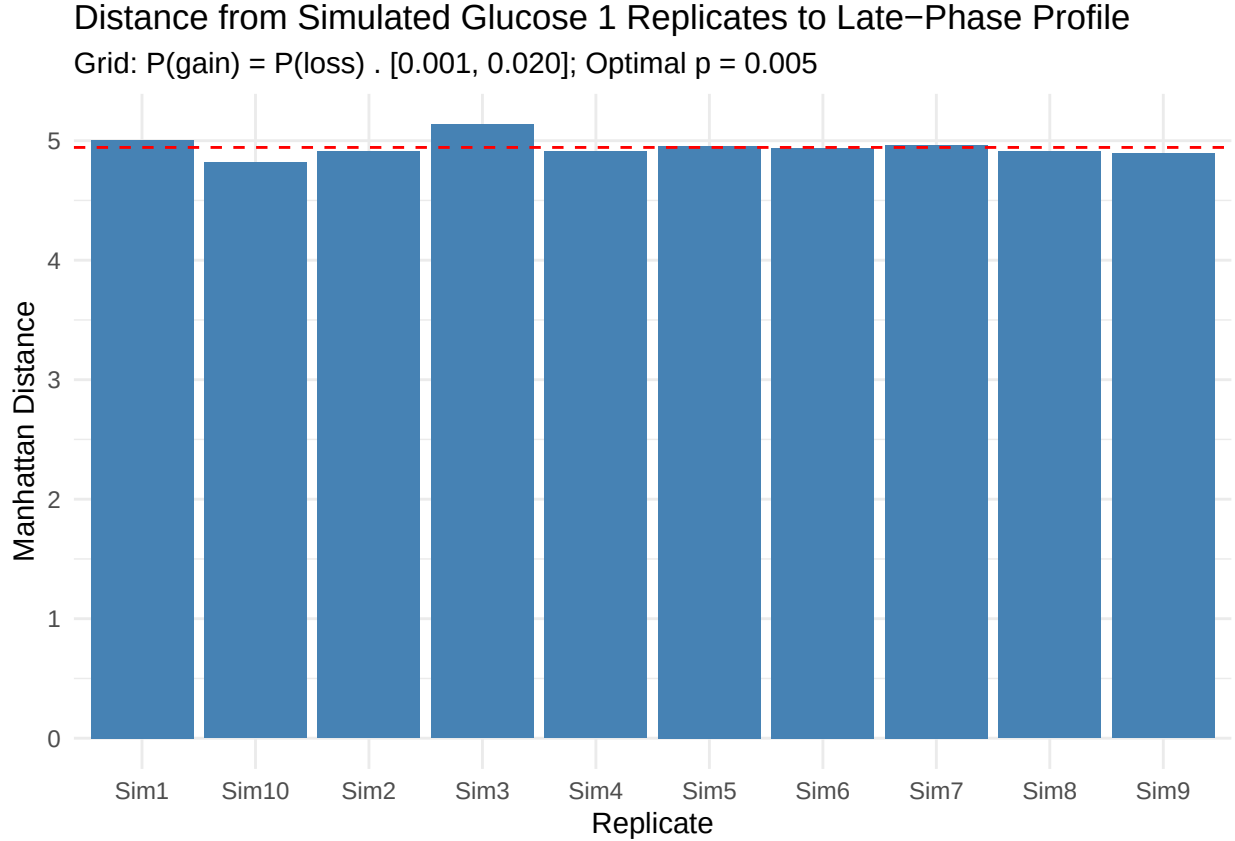


Figure 10: This barplot shows the Manhattan distances between each simulated replicate (Glucose 1, mid-grid search with optimal $p = 0.005$) and the observed late-phase profile. While replicate distances are consistent (~ 4.9 – 5.1), they remain notably higher than those seen under control conditions, indicating poor convergence. This suggests that symmetric gain/loss transitions with $p = 0.005$ are insufficient to reproduce the Glucose 1-induced karyotype evolution, likely due to elevated chromosomal instability not captured by this model.

3.3 Glucose 2 Condition

Option 1: Linear Sequence (for basic grid search)

Table 11: Grid Search Results for Simulated Karyotypes (SNU-668 Glucose 2 Condition)

p	dist
0.001	4.6664
0.002	4.7535
0.003	4.6116
0.004	4.7435
0.005	4.6454
0.006	4.7565
0.007	4.7001
0.008	4.5892
0.009	4.6634

p	dist
0.010	4.6152

Interpretation: The grid search explored a range of symmetric transition probabilities p , where the probabilities of chromosome gain and loss were constrained to be equal ($p_{\text{gain}} = p_{\text{loss}} = p$). This constraint reduced the dimensionality of the parameter space and enabled a more computationally efficient search. For each candidate value of p , we simulated karyotype evolution over 10 discrete time steps (approximately 4 months), starting from early-phase control cells.

Table 12: Optimal Gain and Loss Probabilities Minimizing Distance to Late Glucose Profile

p	dist
0.050	4.4327
0.027	4.5047
0.036	4.5340
0.012	4.5406
0.047	4.5574
0.030	4.5595
0.008	4.5892
0.049	4.6062
0.003	4.6116
0.015	4.6135

Interpretation: Table 12 illustrate top 10 optimized values of p which reflect the most plausible equal gain/loss rates that best reproduce the karyotype transition from early to late Glucose 2 in SNU-668 cells, based on minimizing the Manhattan distance to the observed late-phase profile.

Simulation Replicates and Variability Analysis

To assess the robustness of the model for Glucose 2, we performed replicate simulations using the optimized symmetric transition probability. Variability was quantified using pairwise Manhattan distances between replicate means, capturing stochastic effects. Additionally, three successive refinements of the parameter grid were conducted to ensure that the chosen probability reliably captured the late-phase Glucose 2 karyotype profile, confirming the stability and biological plausibility of the model fit.

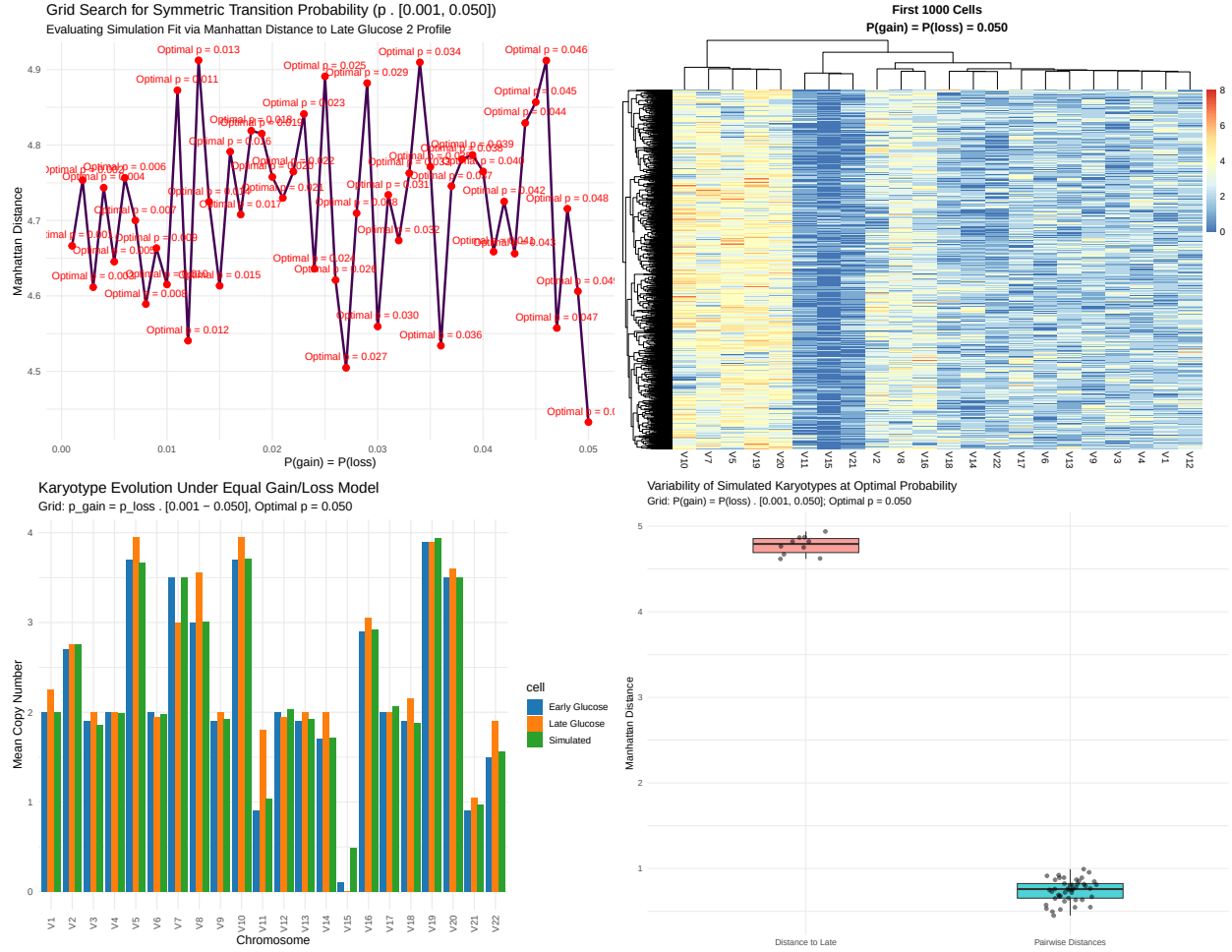


Figure 11: The Glucose 2 linear grid search (optimal $p = 0.050$) reveals that even at the highest symmetric transition rate, simulated karyotypes remain poorly aligned with late-phase profiles (high Manhattan distances). Although the simulation captures elevated chromosomal variability (heatmap, barplot), the boxplot confirms persistent divergence from the late Glucose 2 state. This indicates that Glucose 2 induces strong instability exceeding what symmetric gain/loss alone can explain.

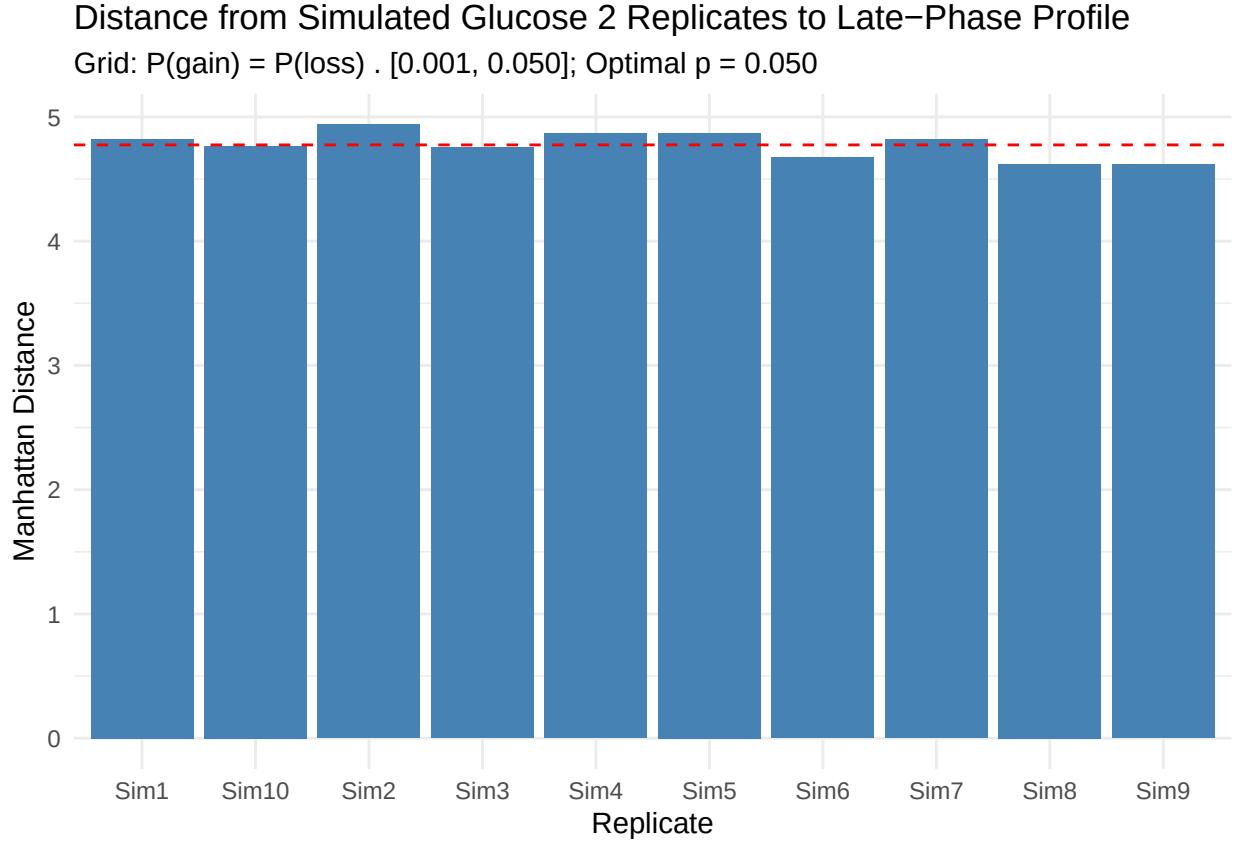


Figure 12: Most replicates fall near the red dashed reference line, indicating consistent simulation performance. While some replicates slightly exceed or fall below the late-phase mean distance, the overall variability is low, and distances remain tightly clustered around ~ 4.9 – 5.0 . This suggests that at high symmetric transition probability ($p=0.050$), the simulations moderately recapitulate the late Glucose 2 karyotype, but improvement over lower p -values is marginal.

Option 3: Mid-Range Balanced Search

Table 13: Grid Search Results for Simulated Karyotypes (SNU-668 Glucose 2 Condition)

p	dist
0.0005	4.6732
0.0010	4.7566
0.0015	4.6325
0.0020	4.6694
0.0025	4.6752
0.0030	4.7816
0.0035	4.6531
0.0040	4.5812
0.0045	4.6483
0.0050	4.5539

Interpretation: The grid search explored a range of symmetric transition probabilities p , where the proba-

bilities of chromosome gain and loss were constrained to be equal ($p_{\text{gain}} = p_{\text{loss}} = p$). This constraint reduced the dimensionality of the parameter space and enabled a more computationally efficient search. For each candidate value of p , we simulated karyotype evolution over 10 discrete time steps (approximately 4 months), starting from early-phase control cells.

Table 14: Optimal Gain and Loss Probabilities Minimizing Distance to Late Glucose Profile

p	dist
0.0050	4.5539
0.0165	4.5606
0.0180	4.5721
0.0040	4.5812
0.0090	4.5853
0.0095	4.5909
0.0135	4.5919
0.0075	4.5966
0.0150	4.6063
0.0060	4.6261

Simulation Replicates and Variability Analysis

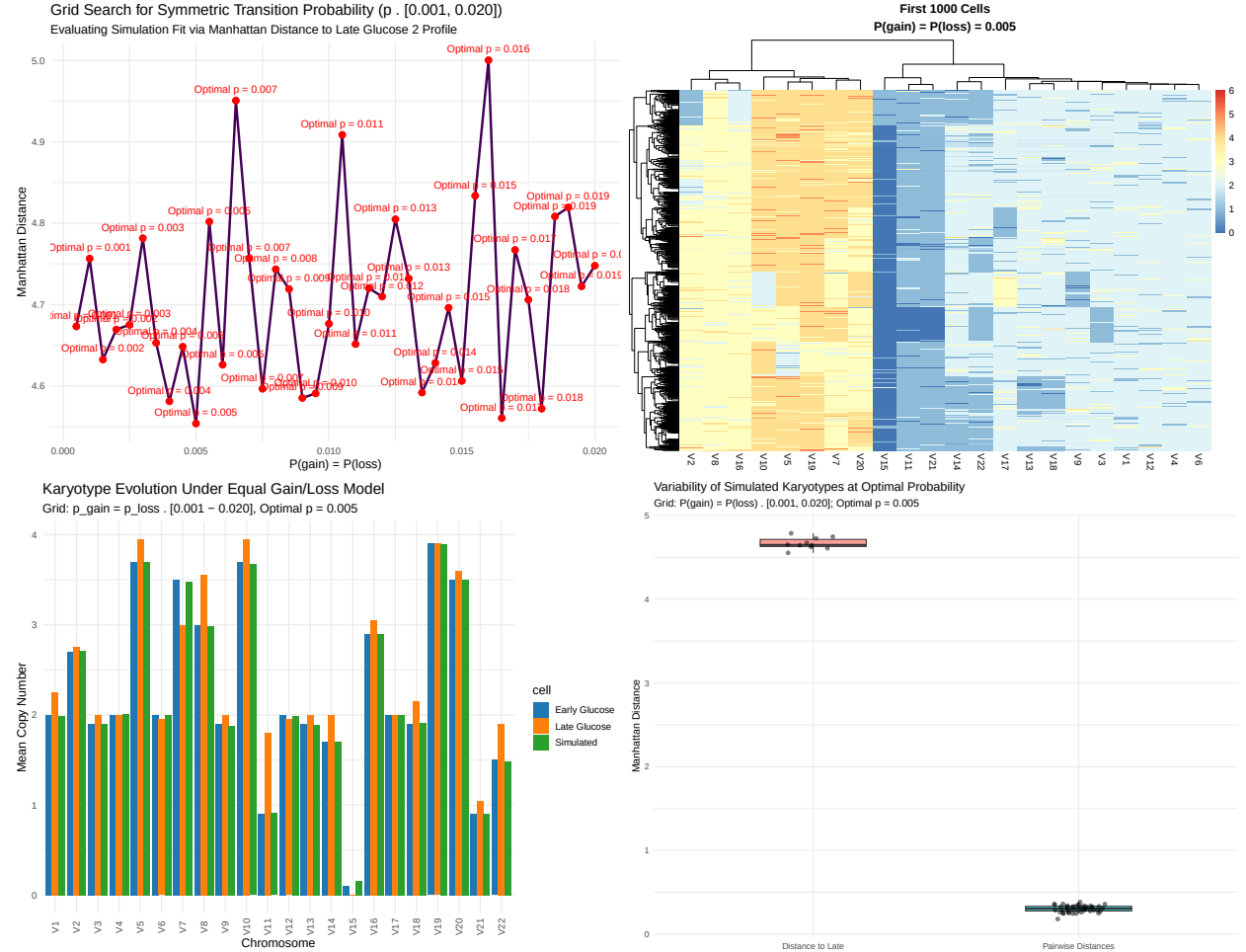


Figure 13: This figure summarizes the results of a mid-range linear grid search ($p \in [0.001, 0.020]$) for Glucose 2, identifying an optimal symmetric gain/loss probability of $p = 0.005$. At this setting, simulated profiles achieved one of the lowest Manhattan distances to the late Glucose 2 profile (~ 4.6), indicating improved fit compared to other grid points. The heatmap of the first 1000 simulated cells reveals structured variability, reflecting key chromosomal features seen in the late-stage profile. The mean copy number comparison shows that the simulated karyotypes closely mirror late Glucose 2, particularly across highly altered chromosomes. Additionally, low pairwise distances and minimal spread in replicate-to-late distances demonstrate consistent simulation behavior across runs. Overall, the optimal parameter yields a biologically plausible and reproducible approximation of late-phase Glucose 2 karyotype evolution.

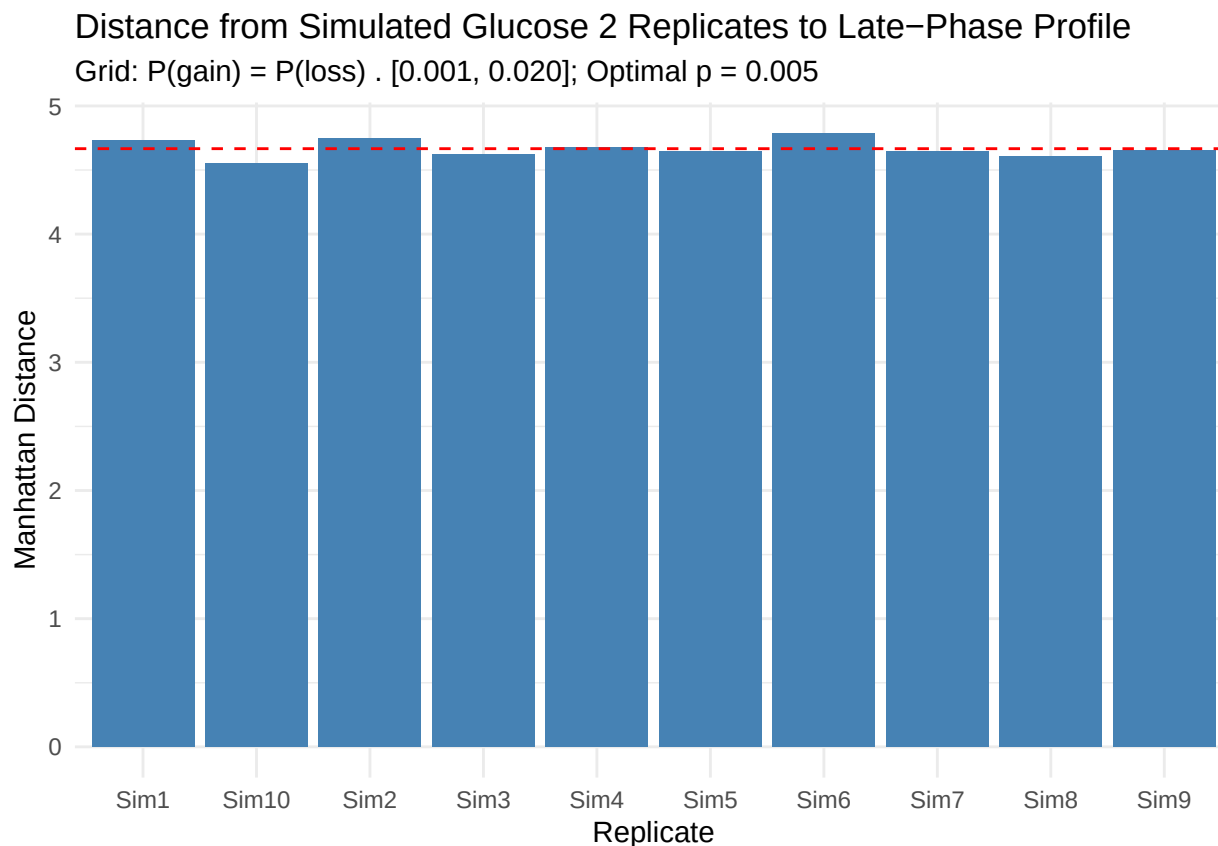


Figure 14: The bar plot displays Manhattan distances between 10 independently simulated Glucose 2 replicates (under optimal symmetric transition probability $p = 0.005$) and the late-phase Glucose 2 profile. Most replicates fall closely around the benchmark distance (indicated by the red dashed line), demonstrating strong consistency in simulation outcomes. This tight clustering confirms that the selected transition probability reliably reproduces late-phase karyotypic patterns across independent runs, supporting model robustness and parameter appropriateness for Glucose 2 under mid-grid conditions.

Discussion

This study employed a discrete-time stochastic model to simulate karyotype evolution in the SNU-668 cancer cell line under three distinct environmental conditions: Control, Glucose 1 deprivation, and Glucose 2 deprivation. The model assumed independent chromosomal gain and loss events across 22 autosomes, constrained by symmetric transition probabilities ($p_{\text{gain}} = p_{\text{loss}} = p$). Grid searches: linear, logarithmic, and mid-range were conducted to identify optimal values of p that minimized the Manhattan distance between simulated average karyotypes and observed late-phase profiles.

Under the Control condition, the model demonstrated strong agreement between simulations and experimental data. Multiple replicates clustered tightly around the late-phase karyotype, and optimal p values consistently yielded low Manhattan distances, indicating the model’s robustness and reliability in capturing chromosomal drift under minimal selective pressure. This suggests that under steady state growth, karyotypic evolution can be reasonably approximated by low-rate symmetric transitions.

For Glucose 1, however, the model exhibited poor convergence regardless of grid type or parameter tuning. Simulated replicates failed to recapitulate the observed late-phase profile, with consistently high Manhattan distances across optimal p values. This discrepancy implies that Glucose 1 treatment induces more complex karyotypic dynamics possibly involving condition specific selection, asymmetry in gain/loss probabilities that are not captured by a Markovian process.

Glucose 2 simulations showed marginally improved convergence compared to Glucose 1 but still underperformed relative to Control. The variability and elevated distance metrics suggest moderate chromosomal instability in response to this condition.

Conclusion

The discrete-time Markov model with symmetric gain/loss transitions performs well under the Control condition, effectively simulating chromosomal evolution over time. However, its explanatory power diminishes under stress induced conditions like Glucose 1 and Glucose 2. These findings underscore the necessity for more nuanced models that account for asymmetric transitions, condition specific selective pressures, and potentially coordinated chromosomal alterations.

Future Directions

Future work should focus on the following extensions to improve model fidelity:

- Introduce asymmetric transition probabilities ($p_{\text{gain}} \neq p_{\text{loss}}$) to reflect directional chromosomal changes under selective stress.
- Incorporate chromosome-specific transition matrices informed by empirical gain/loss frequencies.
- Integrate lineage-tracing data to capture clonal evolution trajectories.
- Explore hybrid models combining mechanistic simulations with machine learning techniques to learn transition structures from data.
- Apply this framework to additional cell lines.